SULIT
Chemistry
Semester 1
Session 2026/2027
2 Hours

NAME :
PRACTICUM :

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ChemWARRIOR

"Master the Concepts, Conquer the Challenge"

No	Marks
1	/ 21
2	/ 10
3	/ 17
4	/ 9
5	/ 9
6	/ 14
TOTAL	/80

%	GRED

INSTRUCTIONS TO CANDIDATE:

This question paper consists of 6 questions.
Answer all questions.
Maximum marks awarded are shown in brackets at the end of each question or section.
The use of electronic calculator is permitted.

RELATIVE ATOMIC MASSES OF SELECTED ELEMENTS

Element	Symbol	Proton number	Relative atomic mass
Aluminum	Al	13	27.0
Antimony	Sb	51	121.8
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Cadmium	Cd	48	112.4
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Cerium	Ce	58	140.1
Caesium	Cs	55	132.9
Chlorine	Cl	17	35.5
Chromium	Cr	24	52.0
Cobalt	Co	27	58.9
Copper	Cu	29	63.6
Fluorine	F	9	19.0
Gold	Au	79	197.0
Helium	He	2	4.0
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Iron	Fe	26	55.9
Krypton	Kr	36	83.8
Lead	Pb	82	207.2
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Mercury	Hg	80	200.6
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Phosphorus	P	15	31.0
Platinum	Pt	78	195.1
Potassium	K	19	39.1
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Scandium	Sc	21	45.0
Selenium	Se	34	79.0
Silicon	Si	14	28.1
Silver	Ag	47	107.9
Sodium	Na	11	23.0
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Tin	Sn	50	118.7
Tungsten	W	74	183.9
Uranium	U	92	238.0
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES

Ionization constant for water at 25°C	K_w	=	1.0×10^{-14} mol^2 dm^{-16}
Molar volume of gases	V_m	=	22.4 dm^3 mol^{-1} at STP
		=	24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum	c	=	3.0×10^8 m s^{-1}
Specific heat of water		=	4.18 kJ kg^{-1} K^{-1}
		=	4.18 J g^{-1} K^{-1}
		=	4.18 J g^{-1} °C^{-1}
Avogadro's number	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant	F	=	9.65 x 10^4 C mol^{-1}
Planck constant	*h*	=	6.6256×10^{-34} J s
Rydberg constant	R_H	=	1.097×10^7 m^{-1}
		=	2.18×10^{-18} J
Molar of gases constant	R	=	8.314 J mol^{-1} K^{-1}
		=	0.08206 L atm mol^{-1} K^{-1}
Density of water at 25°C	ρ	=	1 g cm^{-3}
Freezing point of water		=	0.00 °C
Vapour pressure of water	P_{water}	=	23.8 torr

UNIT AND CONVERSION FACTOR

VOLUME 1 L = 1 dm^3
1mL = 1 cm^3

ENERGY 1 J = 1 kg m^2 s^{-2} = 1 N m = 1×10^7 erg
1 calorie = 4.184 J
1eV = 1.602 x 10^{-19} J

PRESSURE 1 atm = 760 mmHg = 760 torr = 101 325 Pa = 101.325 kPa = 101 325 N m^{-2}

OTHERS 1 Faraday (F) = 96 500 coulumb
1 Newton (N) = 1 kg m s^{-2}

1 (a) **FIGURE 1** shows the mass spectrum of Neon, Ne.

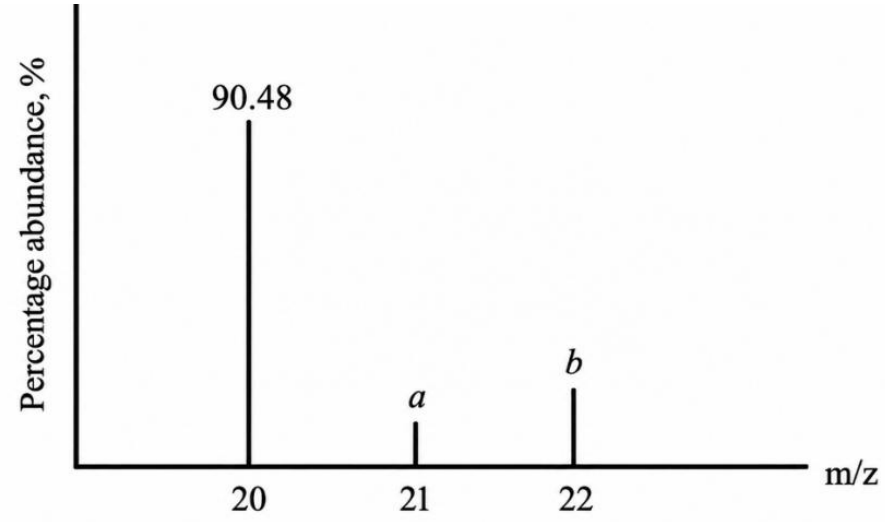


FIGURE 1

Calculate the percentage of abundance of ^{21}Ne and ^{22}Ne if the relative atomic mass of Neon, Ne is 20.18.

[4 marks]

(b) A 10.00 g sample of an organic compound containing carbon, hydrogen, and oxygen is burned in excess oxygen. Complete combustion of the sample produced 14.67 g of CO_2 and 6.00 g of H_2O. Determine the empirical formula of the compound.

[4 marks]

(c) A reagent bottle contains concentrated phosphoric acid, H_3PO_4, labelled as 12.5 molal. The density of the acid solution is 1.685 g mL^{-1}. Calculate the molarity of the acid.

[4 marks]

(d) Balance the following redox equation in basic medium using the ion-electron method:

$$Cr_2O_7^{2-} + I^- \rightarrow Cr^{3+} + I_2$$

[3 marks]

(e) Ammonia, NH_3, is produced by the reaction of nitrogen, N_2 and hydrogen, H_2 gases:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

(i) Calculate the mass of NH_3 produced when 17.0 L of N_2 reacts with 60.0 L of H_2 at room temperature.

(ii) Determine the percentage yield if 18.50 g of NH_3 is collected.

[6 marks]

2 (a) An emission spectrum of a hydrogen atom is observed in the visible region. One of the spectral lines has a frequency of $7.56 \times 10^{14}\ s^{-1}$.

(i) Calculate the wavelength of the line in nanometres (nm).

(ii) Determine the electron transition that produces this spectral line.

[4 marks]

(b) The ionisation of a gaseous hydrogen atom in Lyman series is represented by the equation below:

$$H(g) \rightarrow H^{+}(g) + e^{-}$$

Calculate the ionisation energy of hydrogen in kJ mol^{-1}.

[3 marks]

(c) The quantum numbers of two electrons occupying the highest-energy electron of atom X are given below:

$$(4, 1, 0, +½)\ ,\ (4, 1, -1, +½)$$

(i) Write the electronic configuration of X.

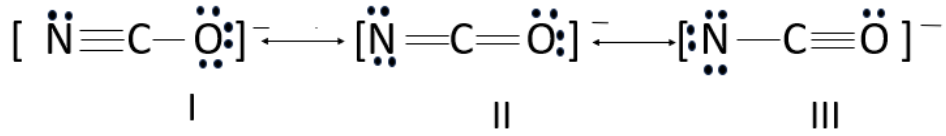
(ii) Draw the orbital diagram for the valence electrons of atom X.

(iii) Draw the shape of the orbital occupied by the 23rd electron in atom X.

[3 marks]

3 (a) (i) Draw the Lewis structure of BF_3 and explain why it does not obey the octet rule.

(ii) The resonance structures of the cyanate ion, NCO^-, are shown below. Determine the formal charge of each atom in each resonance structure and identify the most plausible resonance structure.



I II III

[4 marks]

(b) In a chemistry lab, sulphur difluoride (SF_2) is a specialised, highly reactive main-group chemical primarily used in advanced research to synthesise complex compounds.

(i) Predict the molecular shape and bond angle of SF_2 using the Valence Shell Electron Pair Repulsion (VSEPR) theory.

(ii) Deduce the polarity of the SF_2 molecule.

[4 marks]

(c) The hexachlorophosphonium ion, PCl_6^-, is an octahedral ion commonly found in inorganic compounds.

(i) Show the hybridisation of the P atom in the PCl_6^- ion.

(ii) Draw the overlapping orbitals in PCl_6^- and label the bonds.

[6 marks]

(d) Compare the boiling point of water, H_2O and hydrogen sulphide, H_2S.

[3 marks]

4 (a) The reaction of magnesium metal with hydrochloric acid produces hydrogen gas according to the following balanced chemical equation:

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

If 420 mL of hydrogen gas is collected over water at 22°C and atmospheric pressure of 762 mmHg, calculate the mass of hydrochloric acid reacted.
(The vapor pressure of water at 22°C is 19.8 mmHg)

[5 marks]

(b) **FIGURE 2** shows the phase diagram of compound Z.

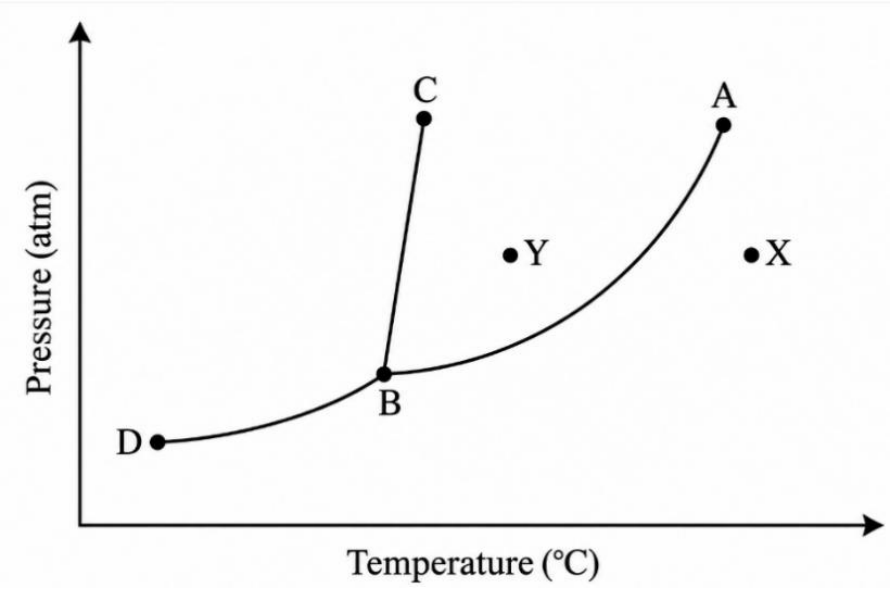


FIGURE 2

(i) Which curve represents the equilibrium between solid and liquid?

(ii) State the phase changes when a sample at point X is cooled at constant pressure until point Y is reached.

(iii) Compare the density of compound Z in the liquid and solid phases. Explain.

[4 marks]

5 (a) Syngas (a mixture of carbon monoxide, CO and hydrogen, H_2 reacts with steam in an industrial vessel to optimize hydrogen yield for fuel cells:

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g) \qquad K_p = 4.0 \text{ at } 500°C$$

A rigid reactor at 500°C is initially charged with a partial pressure of 2.0 atm of carbon monoxide and 3.0 atm of water vapour.

(i) Calculate the partial pressures of carbon monoxide, CO and water vapour respectively once equilibrium is established.

(ii) Determine the total pressure inside the reactor at equilibrium.

[6 marks]

(b) Ammonia is produced industrially via the Haber process according to the following reversible reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The equilibrium constant, K_c, for the reaction is 0.0584 at 498 K. A mixture of nitrogen, hydrogen, and ammonia is introduced into a closed rigid vessel at 498 K. Under these initial conditions, the reaction quotient, Q_c is determined to be 2.98.

(i) Determine the direction of the net reaction to reach equilibrium.

(ii) Explain the effect of adding helium gas, He at constant volume on the equilibrium position.

[3 marks]

6 (a) Pyridine, C_5H_5N, is a weak base. 30.00 mL of 0.60 M pyridine is titrated with 0.50 M hydrochloric acid, HCl solution.
(K_b for $C_5H_5N = 1.7 \times 10^{-9}$)

(i) Calculate the initial pH and final pH of the solution (if 50 mL of titrant was added).

(ii) Qualitatively, predict the pH of the solution at equivalence point. Explain.

[8 marks]

(b) The solubility product of lead iodide, PbI_2 is 8.1×10^{-9} at room temperature. Calculate molar solubility of PbI_2 in water and in 0.2 M $PbSO_4$ solution. Compare the molar solubility values of both PbI_2 in both cases.

[6 marks]